

Directions to pursue

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All of these are ranked approximately in the order that I'd pursue them (which is mostly correlated with immediate importance).

"Necessary" steps

These are mostly concrete central technical steps that directly advance the directions we were already pursuing, or are a fundamental part of our construction and results.

1. **Is *Correct marginalization of hypotheticals* a problematic assumption?**

You were right that given the assumption " $\mathbb{P}$ already knows (within some ϵ) the value $\mathcal{Q}$ assigns to φ " (where $\mathcal{Q}$ is any number representing a market state, and φ any Gödel number of a sentence), your argument goes through and we don't need to explicitly ad hoc what to update on.

But how do we formalize and ensure this? Well, we can consider a formula (or syntactic variable) $M(\cdot, \cdot, \cdot)$ (for *M*arginalization), such that $M(\mathcal{Q}, \varphi) := \mathcal{Q}(\varphi)$ (or instead of equality, to within precision $1/|\mathcal{Q}|$, or some other arbitrary choice). And in our infinite extrapolation of a $\mathbb{Q}_k$ into $\mathbb{P}$, we can ensure $\mathbb{P}$ has the right graph of this function (although note we will have to trash around some existing knowledge if $\mathbb{Q}_k$ already has some wrong opinions about M).

But depending on how we formalize this M , $\mathbb{P}$ will be able to exploit it in more or less ways. And indeed I worry that this is secretly the same as "choosing what to update on" (a choice we had apparently dissolved).¹ For example, as formalized above, if $\mathbb{Q}_k$ believes that "I should check (be updateful about) $M(\mathcal{Q}, \varphi(n))$ ", $\mathbb{P}$ won't be able to exploit this fact, since it won't know the Gödel number of each $\varphi(n)$ (it might know a description of the computation calculating it, but that's not the same as having performed all of these infinitely many computations).

We immediately want to fix this by generalizing M so that it takes parameters automatically, for example $M(\mathcal{Q}, \varphi(x), n) := \mathcal{Q}(\varphi(n))$, so that $\mathbb{Q}_k$ can believe "for all n , I should check $M(\mathcal{Q}, \varphi(x), n)$ ". And I do think this is an especially nice formulation and might be privileged in some way. But a priori it seems like we're still arbitrarily leaving behind further generalizations that would give $\mathbb{P}$ more power. For example, allowing for nesting of this function ($M(\mathcal{Q}, \varphi(x), M(\mathcal{Q}, \phi))$), which amounts to "first check this knowledge, and with that knowledge then check this other one" (that is, 2 meta-levels of general policy selection). This seems like it would give us too general computations (and indeed usually introducing this kind of compositionality makes things explode), but actually I'm not sure that's the case (I don't have a concrete counter-example, and maybe to extract an arbitrary computation a trader needs to write an arbitrarily long nesting so that it gains nothing). It might also be argued that the cases in which this nesting would be required are too weird, but

¹This wouldn't neglect the fact that the discovery of this alternate formulation is surprisingly nice and positive.

actually I think even when Ω 's decision doesn't explicitly depend on "this concrete one of my beliefs on this concrete round" this information could be exploited as a meaningful proxy (that is, it provides information about other more natural things Ω might be thinking about).

- (a) Relatedly, in this framework **which assumptions ensure eventual learning?** Will a mostly uncertain $\mathbb{P}$ lean towards learning? (I think not in general.)
- (b) Relatedly, **how does the 1-step vs N-step difference translate to this new framework?**

2. Formalize and extend "bundles" in LIs

As explained in the document **Reflection principle and Epistemic self-trust**, it seems like we can ensure finite inductor states satisfy *Epistemic self-trust* (up to a finite horizon) by having possible worlds be possible future belief trajectories (up to the finite horizon), putting in the betting table only the "bundles" of possible worlds (probability distributions over possible trajectories) that have the property of Epistemic self-trust, and then letting traders only buy such bundles. The market fix-point (taken as in Universal Inductors with propositionally coherent possible worlds) will be a mixture of these bundles, and so satisfy Epistemic self-trust. Indeed, probabilities over trajectories satisfying Epistemic self-trust are a simplex, since we can break up the conditional into the sum of conditionals of each bundle B (that play the role of hypotheses):

$$\mathbb{Q}_k(\varphi \mid \mathbb{Q}_{k+n}(\varphi) = p) = \sum_B \mathbb{P}(B) \cdot B(\varphi \mid \mathbb{Q}_{k+n}(\varphi) = p) = \sum_B \mathbb{P}(B) \cdot p = p$$

And of course, object-level beliefs about φ are just computed as the marginalization of all these bundles of possible worlds. Also, the other syntactic coherence constraints are now implemented into each of the possible trajectories, so that the marginalized beliefs satisfy them.

Maybe these bundles should also satisfy some form of marginalization between different future stages like $\sum_{p'} p' \cdot \mathbb{P}(\mathbb{Q}_{k+2}(\varphi) = p' \mid \mathbb{Q}(\varphi)_{k+1} = p) = p$.

This same bundle approach can be used to implement different coherence constraints (as long as the probabilities satisfying them form a simplex).

I guess this approach also generalizes to probabilities of probabilities of possible worlds, etc.

There doesn't seem to be a hope (or need) of turning the increasing finite horizon into an actually infinite horizon, since finite LI states have only thought about a finite amount of things.

3. Quantification of conditionals

As explained in my Sep 15 email, we wanted modifications to probability distributions (or things to put on top of them) that allowed us to natively express " $\forall n \mathbb{P}(B(n) \mid A(n)) \approx p$ ", just like we can express " $\forall n A(n) \rightarrow B(n)$ ". It's not clear how necessary this is if the approach in 1. above ends up being satisfactory. My impression is that 1. and 3. are different constructions, and they are both trying to get at "correctly extrapolating small beliefs", but they do so in pretty different ways (and in a sense 3. would seem like a more fundamental and less ad hoc modification of how probabilities work). We have several ideas in this direction:

- (a) **Extrapolating traders:** As in my Sep 15 email, by making explicit what we exactly need (and its relation to the Reference Class problem), I was naturally led to the idea of extrapolating conditionals by fixing the traders we already have (without implementing any new ones, as happens in every round of a LI) and asking that limited market for opinions on different situations. This is like becoming updateless about new possible hypotheses we haven't yet considered (black swans, etc.), but continuing to use the already present heuristics. Technically

speaking, this might play the role of "a more native extrapolation" / "a more natural boundary to set between the updateless and the updateful". But we seem to still need an additional mechanism, since this is purely epistemic and so would in some cases be updateful when that's not the best play. And also, if we wanted to go in the direction of "realist logical uncertainty", it doesn't solve the problem of traders being implemented by real-world computations.

- (b) **Introducing a symbol for conditionals:** Natural thing to try. Possibly related to your solution to counterfactuals? (Introducing a symbol, and making it coincide with conditional in all real instances.)
- (c) **Other implementations of conditional stability:** That is, other ways to look at past conditionals and extrapolate which ones should be preserved. But it seems like the "true name" of this is (a) above.

4. Running out of opinion juice

We want our agent to stop having strong opinions (and start updating) after some time. This seems to require two steps:

- (a) First, the **opinions do actually have to fade** in some way. It seems this is blocked by Universal Instantiation. You mentioned something like, if $\mathbb{P}$ "starts doubting" the universally quantified formula, it will also doubt (and maybe be willing to update on) each of the particular instances of that universal. But I think this is not true, because "starts doubting" just means " $\mathbb{P}$ was uncertain (or willing to update) from the start". So it was also willing to update on each of the particular instances from the start (due to our coherence properties). By further pursuing some of the above directions, we might find a natural or privileged way to make opinions fade. Otherwise, our only path will be to implement it ad hoc in $\mathbb{P}$, as we have always done.
- (b) Second, **this fading does have to imply more updating**. If 1.(a) doesn't work out as wanted, we might find the only way to ensure this is again implement a modification as in 3.

There might be other modifications that don't naturally break down into these two steps. We mentioned modifying traders (but the first idea you had was just "forcing less confidence on some universals", which sounds very unnatural).

5. Think more about impossibility results

In particular, prove some things about average behavior across trees, as in my document **Impossibility results**. The possibility of much more interesting results would open if we found other notions of optimality as in 7., or a realist picture of logical uncertainty as in 8.

6. Fancier UDT1.1

As explained in the document **Getting UDT1.1 (Fixed version)**, the proposal there is increasing a finite policy horizon, but maybe we'd be able to take an actually infinite horizon by conditioning $\mathbb{P}$ on an infinite conjunction of policy points. This of course means taking the limit of finite conjunctions. This doesn't always converge, but under our assumptions about time-penalized utility (or at least not unbounded utility in any infinite branch or finite level) it always does, since arbitrarily far-off points can only have an arbitrarily small impact on utility (this is just basically applying a time penalty so hard that the situation can be arbitrarily well-approximated by a finite tree).

Live threads

These are bigger, more open-ended and conceptual topics that, while still related to our work, might take it in new directions and yield some radically new perspectives.

7. Notions of optimality

As I discussed in my presentation and elsewhere, we cannot motivate "what full updatelessness misses" from within the Expected Value Maximization framework (from its subjective perspective, it's just doing EVM, and we can't blame it for that). We need to formalize mathematically what it is from complete updatelessness that "feels wrong" to humans, because of being too risky, or because "obviously you will be missing a lot of information, and this will on average play out against you". What twisted and value-laden notion of "on average" are we using here?

It might be that something quite natural arises, of the form: When we look at these agents from the outside (treating them as algorithms we can argmax over), according to some way of taking the average that is especially natural, it turns out it is pragmatically privileged to eventually learn (although this is of course already theory-laden on the definition of average).

But even something less "motivated" and more "this formalization / way of taking the average just captures our intuitions, even if we don't know why for now" would work.

8. Realist logical counterfactuals

We've talked about this a lot, and now we both agree that it's worthwhile to look at this from an evolutionary / Selection Theorems perspective. Maybe there is no "objectively realist" notion (it's all mind-dependent), but all minds (for efficiency reasons a la natural abstractions, or other reasons) converge on the same kind of cognitive architecture to deal with logic, and so we can communicate easily and robustly logical counterfactuals (even with aliens and AIs), and it's something pragmatically favored. This amounts to "being updateful about the fact that in this universe most minds look like this", but we're probably willing to do that (this might be the "un-dissolvable ground"). This might be related to how updatelessness and communication trade against each other (Critch's democratic RL amounts to communal updating even if the individuals are updateless, also playing for the world you want vs best-responding in the world you are in, etc.), so that this last "un-dissolved ground" enables efficient communication.

9. LIs' shortcomings (agent's perspective)

LIs don't behave that well epistemically with universal sentences. Might this have bad consequences for the performance of our agents? Does the Pi1-Pi2 impossibility result have consequences for decision-theoretic performance of logically uncertain algorithms? Of course, maybe we find fixing some of these shortcomings would amount to "finding something like LIs but better", which sounds hard.

10. LIs' shortcomings (predictor's perspective)

Either due to the above shortcomings, or even due to epistemically desirable features of LIs, might complex enough algorithms **A** be able to do something that intuitively looks like "tricking the LI"? A first idea is explored in the document **Problem or opportunity: LIs do sometimes get tricked**. It seems for now like tricks can only happen when **A**'s probabilities of taking actions, and thus also its payoff, don't converge. If this were true, it wouldn't be as pathological, since it's less clear that a payoff infinitely flailing around (even if it reaches arbitrarily high numbers) is an improvement for the agent. Of course, in a more realistic situation, this would be very worrisome (a powerful enough agent could wait until a very high payoff and then stop playing). But it's

not clear what the learning-theoretic interpretation of this should be, and this seems already like Computational Complexity territory, as in 11.

In the drawer

These are interesting ideas that have not been brought to life yet.

11. **Introducing Computational Complexity**

There might be interesting things to say about the complexity of our current setups. And maybe adding computational constraints on top of the learning-theoretic constraints we already have would yield something surprisingly nice or natural.

But the original motivator for introducing Computational Complexity, as expressed in the document **Argmaxing our strategy**, was a different one, which can be seen as alternative to our current framing. In our current framing, we embrace embeddedness, and this brings us into a lot of problems with the arbitrariness of subjective logical uncertainty. We want to say objective things, or find different notions of optimality as in 7., but what can we say other than "Yes, from this agents' perspective (that believes this crazy correlation), the best EVM action was to shoot itself in the foot"?

The alternative is to look at our agent from the outside (not subjective), or even ensure that Omega's LI only looks at actions (so that our algorithm is choosing action policies "from the outside"). This breaks real embeddedness, and turns "just argmax over programs / action policies" into the obvious computable solution, satisfying all notions of optimality we could dream of. But maybe if we introduce computational constraints (so that the brute-force search for argmax is not feasible) this becomes interesting again. And maybe some dynamics arise that are the pragmatic analogue to embeddedness or updatelessness. As in, "I could indeed just update on everything if I had the computing power, but since I don't, it's best for me to act as if I were updateless about some things". But this might collapse into trivialities (if you don't have the computing power, you don't get to choose whether you're updateless, you just automatically are).

12. **Connection to EDT + Anthropics**

We discussed how, as per Jessica's and Caspar's results, non-updateless decision theories can act updateless due to simulations. I have the cached memory that this still didn't give us everything we wanted (even in perfect simulation situations), but I might be wrong. But even if that wasn't the case, when we introduce non-perfect simulations, we find again the problem of "what effect does this computation have in this other one" (propagating logical information), as in 8.

13. **Game Theory of all this Learning Theory**

There are many questions about Game Theory we could put our framework against, to see how it responds. Wei had some.

14. **Learning Theory of Nisan's Hierarchical Game Theory**

15. **Finite State Machines converging to good equilibria in iterated games**

A paper exists on this (although pretty different direction, much more computational complexity). Is this a particular case of existing Delegated Game-Playing literature?

We might want our automata to have two parts, one representing beliefs and other actions?

Similarly, we can evaluate "actions given beliefs", but also "how lucky was I to get these beliefs" (according to some different external objective metric).